

Blessings Mambwe

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(GMT+2) Lusaka, Zambia

SKILLS

- **Languages**: Dart ◇ Python ◇ Javascript
- **Frameworks**: Flutter ◇ Docker ◇ FastAPI ◇ Pytorch/Tensorflow ◇ Google Cloud ◇ AWS ◇ SQL ◇ Tableau ◇ PowerBI

EXPERIENCE

- **Stanbic Bank** Aug 2025 – Present | Hybrid
AI & Automation Engineer
 - ▷ Automated MT940 statement processing using SQL, Python, and Power Automate, enabling faster reconciliation and supporting a projected \$1.3M increase in liability capacity.
 - ▷ Built a secure AI-powered HR conversational interface with governance, compliance, and data-leakage controls for safe enterprise knowledge access.
 - ▷ Designed a collateral management system backed by SQL, automating client collateral allocation, report generation, and scheduled email distribution.
 - ▷ Led a data science upskilling initiative in collaboration with Zindi, strengthening local AI talent and co-driving Zambia's largest AI and data science event.
- **MTN** Jan 2025 – August 2025 | Remote
Digital Transformation Analyst
 - ▷ Developed a full-stack gamified aptitude test platform with anti-LLM safeguards and real-time scoring <https://limitless-4f89b.web.app>, helping HR teams save thousands of dollars in screening costs.
 - ▷ Led project management for the myMTN app, collaborating with stakeholders, IT, Group, and third parties to resolve technical issues, introduce new features and drive revenue growth.
 - ▷ Analyzed call center data to identify common queries, redirecting 25% of traffic to the myMTN app, saving \$50,000.
 - ▷ Analyzed app user data to segment customers by age, location, and product, delivering targeted SMS prompts that boosted app activity by 9%.
 - ▷ Managed the Facebook Mobile Center portal to track product performance, collaborating with Facebook and IT to enhance API success rates, leading to increased revenue.
- **MarcTina Consultancy** Jan 2024 – Dec | Remote
Data Scientist
 - ▷ Utilized large language models to analyze and derive insights from text data.
 - ▷ Developed a Python-based resume-processing web app using LangChain APIs, deployed on Google Cloud for real-time analysis of CVs and job details, available at <https://resu-ai.web.app>
 - ▷ Led training sessions on data architecture, SQL, data warehouses, and industry best practices.
- **Speak Life AI** Mar 2024 – Present | Remote
AI Engineer
 - ▷ Independently developed a cross-platform faith-based web app for generating biographies, devotions, and scripture searches from the Bible based on user prompts, available at <https://speaklifeai.com>
 - ▷ Implemented robust data collection, access procedures, and change processes, maintaining a comprehensive data dictionary for the project.
- **ML Collective** Jun 2020 – Present | Remote
Machine Learning Researcher
 - ▷ Successfully deployed the most efficient object detection model, Efficient Detection, to detect airplanes from satellite images using Python and PyTorch.

- ▷ Collaborated and communicated perfectly with 2 researchers to co-author and submit a blog post to ICLR comparing Deepmind's mechGraphNets model to molecular particle simulators. I simulated flag dynamics, bash scripted to download Blender and created renders using cloud GPU on Google Colab (Linux). See [\[report\]](#).
- ▷ Developed PRISM, a permutation-based neural architecture search algorithm with provable convergence guarantees and polynomial runtime complexity; authored a comprehensive research summary and presented the underlying neuro-evolutionary theory before experts, see [\[presentation\]](#).

• IO Technologies

Apr 2023 – Jun 2023 | Remote

Fullstack Software Developer - Contract

- ▷ Independently developed a sophisticated vehicle tracking system responsible for notifying users of driving violations, including overspeeding and night driving infractions to adhere to traffic regulations.
- ▷ Designed an efficient algorithm in Python to detect instances of violations from hundreds of vehicles by calculating time, speed, and other parameters per API request. Updated the database and leveraged multithreading and the Outlook API to dispatch violation alerts instantly to emails.
- ▷ Leveraged AWS Serverless lambda to seamlessly present real-time vehicle information, facilitating remote debugging for enhanced efficiency.

• Konkola Copper Mines

Oct 2022 – May 2023 | Chingola, Zambia

Mechanical Engineer Trainee

- ▷ Developed Python-based predictive analytics model for pump vibration data, enabling accurate failure forecasting and facilitating proactive maintenance strategies, resulting in improved operational readiness.
- ▷ Led the implementation of YOLOv8, an advanced image classification model, for predictive maintenance, resulting in a \$300k cost-effective solution that significantly reduced hazards and downtime.
- ▷ Analyzed SCADA system data to diagnose pump failures and implemented a systematic review process of artisan reports, ensuring up-to-date maintenance protocols and enhancing overall system reliability.

• Afri-Meta

Jun 2022 – Sep 2022 | Lusaka, Zambia

Fullstack Software Developer Intern

- ▷ Spearheaded a cross-functional team of 4 in developing an advanced food classification system, leveraging data engineering principles to create a robust ETL pipeline and deploying scalable machine learning models on Google Cloud Platform.
- ▷ Engineered and implemented a high-performance Normalizer-Free 'EffNetV2' machine learning model, utilizing Python and PyTorch for data preprocessing and model training. Achieved 93% accuracy with 0.8-second inference time, demonstrating expertise in optimizing large-scale data processing systems.
- ▷ Architected and executed an end-to-end MLOps pipeline on Google Cloud Engine, incorporating Docker for containerization and custom Linux Bash scripts for automation. Implemented data versioning and model monitoring to ensure consistent performance and facilitate iterative improvements. Demo: [\[link\]](#).

ENTREPRENEURSHIP & PRODUCTS

- **EVE — AI Financial Intelligence Assistant** [🔗](#): Founded and independently engineered a cross-platform AI financial intelligence app: users track stocks, crypto, and forex on interactive OHLC charts and converse with a voice-enabled AI assistant grounded in live market data. Built with Flutter, Python FastAPI on Google Cloud Run, and Firebase; launched to production on the Google Play Store and web with CI/CD, in-app payments, and OTP authentication.

OPEN-SOURCE ENGAGEMENT AND INDEPENDENT RESEARCH PROJECTS

- **LabLab AI LLM Hackathon**: Developed full-stack healthcare recommendation web application integrating LLM APIs for personalized fitness and medication recommendations based on patient conditions, featuring automated data pipelines and model deployment available at [Lablab.ai](#).
- **Credit Risk Prediction**: Developed credit risk prediction model using random forest classifier achieving 93% ROC AUC and 92% recall/accuracy. Enhanced model performance through feature engineering of temporal patterns and loan characteristics, with cross-validation across multiple algorithms.
- **Absa Bank Cybersecurity Hackathon - 2nd Place**: Developed a real-time cyber threat intelligence platform using Flutter (Dart), Python, and NoSQL, enabling secure data aggregation, visualization, and reporting, guided by Bank of Zambia and ZICTA on regulatory-compliant cyber resilience.

- **MTN Mobile Money Hackathon:** Built and deployed a customer support LLM agent using RAG on MTN data with Python, LangChain, and the ChatGPT API, enforcing domain-safe prompt controls and delivering a Flutter-based UI hosted on Google Cloud.
- **Estimating Solubility of Chemical Elements using GraphSAGE:** Used Python with PyTorch to deploy the GraphSAGE model to classify proteins from big datasets of proteins. It archived accuracy of over 80%, wrote a report which included essential graph algorithms. See [\[report\]](#).

EDUCATION

- **TRI // DeepMind AI Research Training:** 2026 - Present ◦ Research paper in progress: *LLM Latent Probing for Toxicity Detection*
 - **Professional Training:** ◦ Enterprise API Training – Stanbic Bank ◦ AI Governance – PECB
 - **Independent Study:** 2019 - 2021 ◦ A Matrix Algebra Approach to Artificial Intelligence ◦ Approximation Theory and Algorithms for Data Analysis ◦ Data Structures, Algorithms and System Design
 - **Manchester Metropolitan University :** 2015 - 2019 ◦ BEng (Hons) Mechanical Engineering | Manchester, UK
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- **Awards 🏆 :** ◦ Carbon Literacy ◦ Hour of Code 🌟 | **Hobbies:** *Chess Sound Design 🎵 Music Production 🎧*